GPSC CBRT · HOMOEOPATHIC PHYSICIAN · CORE

Topic 5: General & Applied Human Biology

Complete notes with pictures, memory hooks, rapid revision and practice MCQs.

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Topic 5: General & Applied Human Biology

How to use this guide

Topic 5 gives about 2 of the 25 core marks, and almost every question is a one-line fact: a normal value, a nerve that supplies a muscle, a hormone and its gland. Learn the tables and the labelled diagrams first; the prose only links them.

Syllabus (Advt 06/2026, Topic 5). The sections below follow it in order:

Syllabus heading	What it covers	Section here
Applied neuro-musculoskeletal anatomy	Clinically relevant osteology, joint classification, brain localization, peripheral nerve pathways	5.1
Cardiovascular, respiratory and renal physiology	Cardiac cycle, ECG, regulation of blood pressure, mechanics of respiration, renal clearance	5.2a–c
Gastrointestinal and endocrine systems	Digestion, absorption, hepatic functions, hypothalamic-pituitary axis, feedback loops	5.3a–b
Body fluids and homeostasis	Fluid-electrolyte balance, acid-base balance, clinical biochemistry (enzymes, lipid profile, blood glucose)	5.4a–b

- **Standard books used.** Guyton and Hall, *Textbook of Medical Physiology*; Ganong, *Review of Medical Physiology*; A. K. Jain, *Textbook of Physiology*; B. D. Chaurasia, *Human Anatomy* (vols 1–4); Vishram Singh, *Textbook of Anatomy*; Vasudevan and Sreekumari, *Textbook of Biochemistry*; Satyanarayana, *Biochemistry*.
- These are the anatomy, physiology and biochemistry books prescribed in the BHMS first-year (NCH/NCISM) course.
- Normal values follow these books; laboratories differ slightly, so learn the range, not the last decimal.

Plan to 1 November (12 days). Two sections a day for days 1–5 (diagrams first, then tables); day 6 the rapid-revision sheet and the practice MCQs; after that, 15 minutes a day on the rapid-revision sheet alongside your mock tests.

5.1 Applied neuro-musculoskeletal anatomy

Examiners ask four kinds of question here: which bone or joint (with its type), which part of the brain does what, which nerve supplies a muscle, and which deformity follows a nerve injury.

Clinically relevant osteology

Bone or landmark	High-yield fact
Clavicle	First bone to ossify (5th–6th week of intrauterine life); ossifies in membrane; commonest bone to fracture, at the junction of its middle and lateral thirds
Scaphoid	Commonest carpal bone fractured (fall on outstretched hand); tenderness in the anatomical snuffbox; risk of avascular necrosis of the proximal fragment
Surgical neck of humerus	Fracture injures the axillary nerve
Shaft of humerus (spiral groove)	Fracture injures the radial nerve: wrist drop
Medial epicondyle of humerus	Ulnar nerve runs behind it (the "funny bone")
Neck of femur	Fracture in the elderly; risk of avascular necrosis of the head (blood supply via retinacular arteries)
Neck of fibula	Common peroneal nerve winds round it: foot drop
Femur	Longest and strongest bone
Stapes	Smallest bone
Pisiform	Sesamoid bone in flexor carpi ulnaris; patella is the largest sesamoid bone
Atlas (C1)	No body, no spine; atlanto-occipital joint allows nodding ("yes")
Axis (C2)	Dens (odontoid process); atlanto-axial joint allows rotation ("no")
Vertebra prominens	C7, first prominent palpable spinous process
Sternal angle (of Louis)	Level of the 2nd costal cartilage and the T4–T5 disc; used to count ribs
Spinal cord	Ends at the lower border of L1 in adults (L3 at birth); lumbar puncture is done at L3–L4

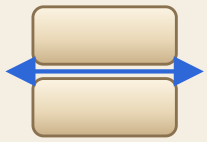
Adults have 206 bones; a newborn about 270–300, many of which fuse.

Joint classification

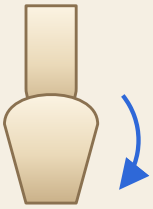
Type	Sub-type	Example
Fibrous	Suture	Skull sutures
Fibrous	Syndesmosis	Inferior tibiofibular joint
Fibrous	Gomphosis	Tooth in its socket
Primary cartilaginous (synchondrosis, hyaline)	Temporary	Epiphyseal plate; 1st chondrosternal joint
Secondary cartilaginous (symphysis, fibrocartilage)	Midline, permanent	Pubic symphysis; intervertebral discs; manubriosternal joint
Synovial	Plane	Intercarpal, acromioclavicular
Synovial	Hinge	Elbow (humero-ulnar), interphalangeal, ankle
Synovial	Pivot	Superior radio-ulnar; atlanto-axial (median)
Synovial	Condylar (bicondylar)	Knee, temporomandibular
Synovial	Ellipsoid	Wrist (radiocarpal), metacarpophalangeal
Synovial	Saddle (sellar)	First carpometacarpal joint of the thumb
Synovial	Ball and socket	Shoulder, hip

Synovial joints by shape

The shape of the bone ends decides the movement



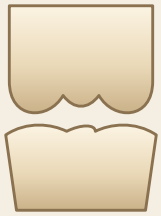
Plane
Intercarpal, acromioclavicular



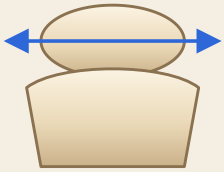
Hinge
Elbow, ankle, interphalangeal



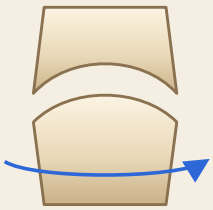
Pivot
Superior radio-ulnar; atlanto-axial



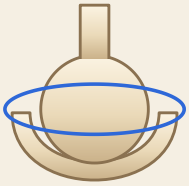
Condylar
Knee, temporomandibular



Ellipsoid
Wrist (radiocarpal), MCP joints



Saddle
1st carpometacarpal (thumb)



Ball and socket
Shoulder, hip

Not synovial

Fibrous: sutures, syndesmosis, gomphosis (tooth in socket)

Primary cartilaginous: epiphyseal plate

Secondary: pubic symphysis, discs

Hook: "Saddle for the thumb, ball for the limb roots." Blue arrows show the movement each shape allows.

Synovial joints by shape · the shape of the bone ends decides the movement

Memory hook: "Saddle for the thumb, ball for the limb roots." The thumb's saddle joint allows opposition; the shoulder and hip are the only ball-and-socket joints in the limbs. The shoulder is the most mobile and most often dislocated joint (anterior dislocation, axillary nerve at risk).

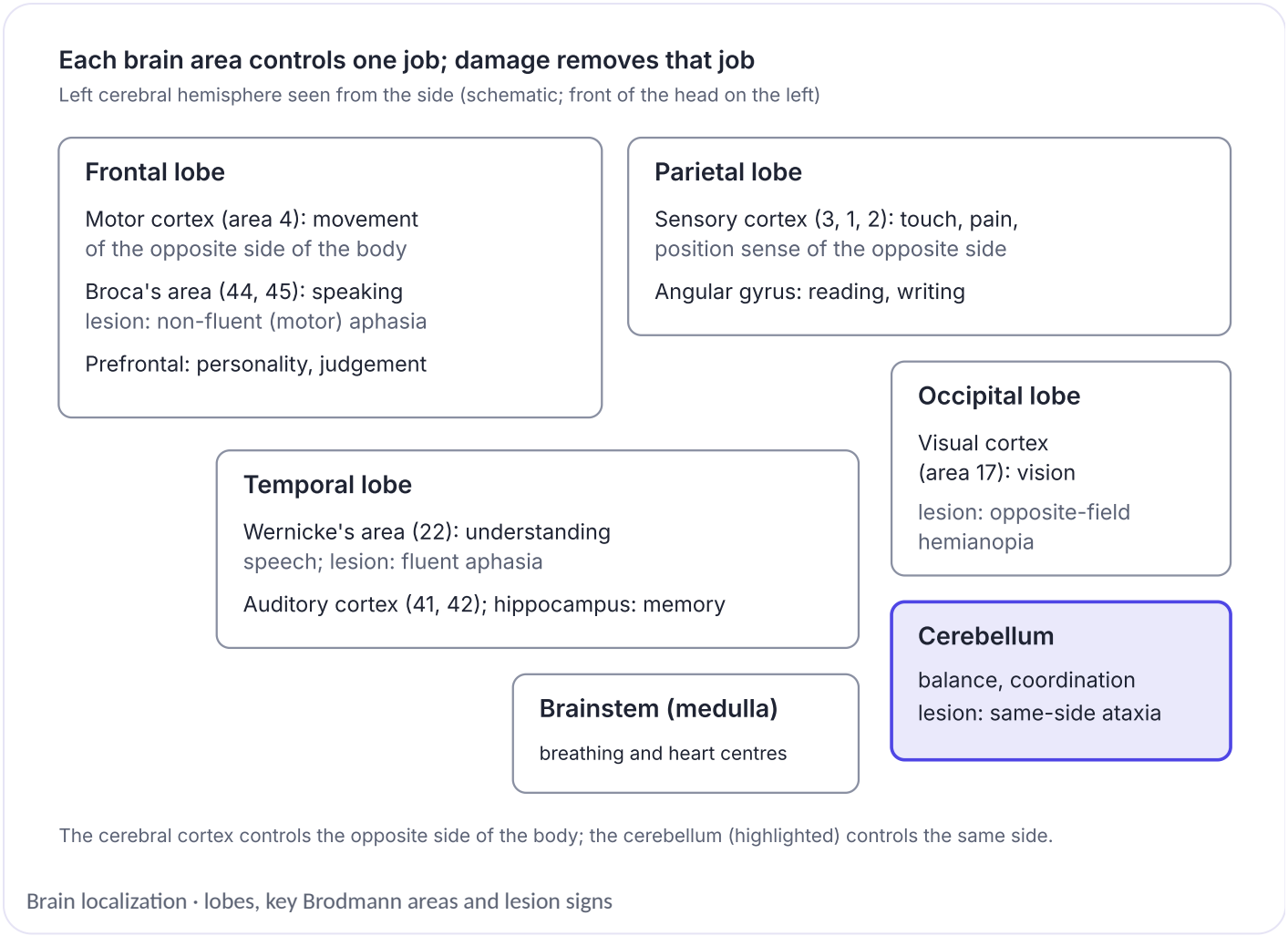
Brain localization

The labelled map below shows which area does what; a lesion causes loss of that function.

Area (Brodmann)	Function	Lesion
Primary motor cortex, precentral gyrus (4)	Voluntary movement of the opposite half of the body	Contralateral upper motor neuron weakness
Premotor area (6)	Planning of movement	Apraxia
Broca's area, inferior frontal gyrus, dominant side (44, 45)	Motor speech	Expressive (non-fluent) aphasia: understands, cannot speak fluently

Area (Brodmann)	Function	Lesion
Prefrontal cortex	Personality, judgement	Disinhibition, personality change
Primary sensory cortex, postcentral gyrus (3, 1, 2)	Touch, pain, position of the opposite side	Contralateral sensory loss
Wernicke's area, superior temporal gyrus, dominant side (22)	Understanding speech	Receptive (fluent) aphasia: fluent but meaningless speech
Primary auditory cortex, Heschl's gyrus (41, 42)	Hearing	Little deafness (bilateral input)
Hippocampus (medial temporal lobe)	New memory	Anterograde amnesia
Primary visual cortex, calcarine sulcus, occipital lobe (17)	Vision	Contralateral homonymous hemianopia with macular sparing
Cerebellum	Coordination, balance, tone	Ipsilateral ataxia, intention tremor, nystagmus, dysdiadochokinesia
Basal ganglia (substantia nigra, corpus striatum)	Initiation and smoothness of movement	Parkinsonism (dopamine loss in the substantia nigra)
Hypothalamus	Temperature, hunger, thirst, pituitary control	Diabetes insipidus, temperature and appetite disorders
Medulla oblongata	Respiratory and cardiovascular centres	Death if destroyed ("vital centres")
Internal capsule (posterior limb)	Corticospinal and sensory fibres	Contralateral hemiplegia; commonest site of stroke

Memory hook: "B for Broca, B for Broken speech; W for Wernicke, W for Word salad." Broca is in front (frontal, motor), Wernicke behind (temporal, sensory).



Brain localization · lobes, key Brodmann areas and lesion signs

Read it as a map: frontal = movement and speech output, parietal = sensation, temporal = hearing, understanding and memory, occipital = vision; the cerebellum is the one part that works on the same side.

Peripheral nerve pathways and injuries

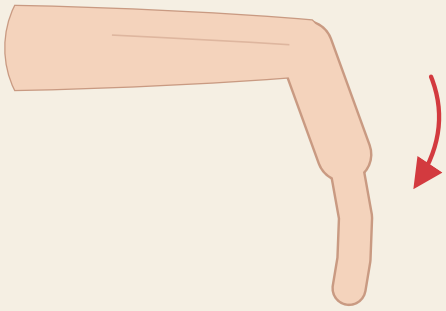
Brachial plexus. Roots C5–T1, then trunks, divisions, cords, branches ("Read That Damn Cadaver Book"). The cords are named for their relation to the axillary artery: lateral, medial and posterior.

Nerve (root value)	Supplies (key muscles)	Injury and sign
Axillary (C5, C6)	Deltoid, teres minor	Surgical-neck fracture or shoulder dislocation; loss of abduction 15–90°; numb "regimental badge" patch
Musculocutaneous (C5–C7)	Biceps, brachialis, coracobrachialis	Weak elbow flexion and supination
Radial (C5–T1)	Extensors of arm and forearm (triceps, wrist and finger extensors)	Spiral-groove fracture or "Saturday night palsy"; **wrist drop**

Nerve (root value)	Supplies (key muscles)	Injury and sign
Median (C5–T1)	Most forearm flexors; thenar muscles; lateral two lumbricals	Carpal tunnel syndrome; **ape-thumb hand**; at the elbow: **hand of benediction** on trying to make a fist
Ulnar (C8, T1)	Most intrinsic hand muscles; flexor carpi ulnaris	Medial-epicondyle injury; **claw hand** (ring and little fingers); Froment's sign
Long thoracic (C5–C7)	Serratus anterior	**Winging of the scapula**
Upper trunk (C5, C6)	—	**Erb's palsy**: "waiter's tip" position (birth injury, fall on shoulder)
Lower trunk (C8, T1)	—	**Klumpke's palsy**: claw hand, may have Horner's syndrome

Nerve injuries you can see in the hand

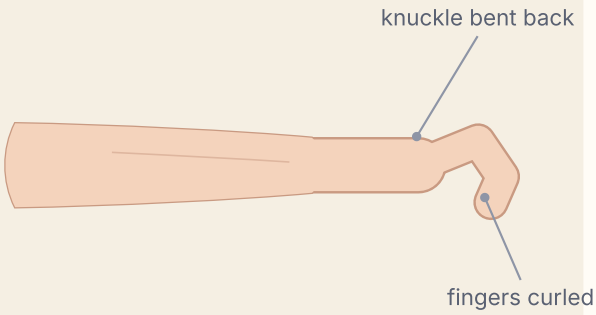
Radial drops the wrist, ulnar claws, median blesses or flattens the thumb



Wrist drop

Radial nerve (C5–T1)

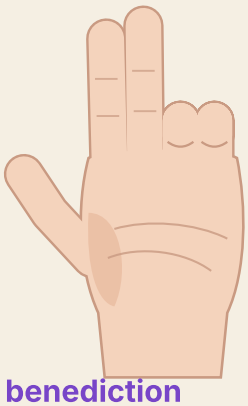
Spiral-groove fracture of the humerus or "Saturday night palsy"
Wrist and finger extensors fail



Claw hand

Ulnar nerve (C8, T1)

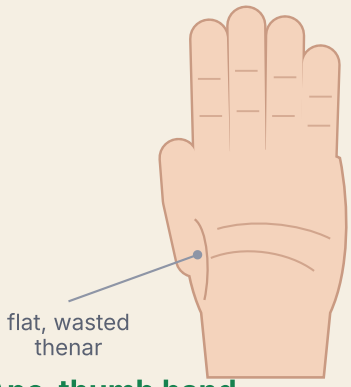
Ring and little fingers: knuckles bent back, finger joints curled
Medial epicondyle injury; Froment's sign



Hand of benediction

Median nerve at the elbow

On trying to make a fist, the index and middle fingers stay straight



Ape-thumb hand

Median nerve at the wrist

Carpal tunnel syndrome: thenar muscles waste, the thumb lies flat in line with the fingers

Nerve injuries in the hand · wrist drop, claw hand, benediction, ape thumb

Lower-limb nerve	Supplies	Injury and sign
Femoral (L2–L4)	Quadriceps	Lost knee extension and knee jerk
Obturator (L2–L4)	Adductors of thigh	Weak adduction
Sciatic (L4–S3)	Hamstrings and all muscles below the knee	Thickest and longest nerve; injured by a wrongly placed gluteal injection (give it in the upper outer quadrant)
Common peroneal (L4–S2)	Dorsiflexors and evertors of the foot	Fibular-neck injury; **foot drop**, high-stepping gait

Lower-limb nerve	Supplies	Injury and sign
Tibial (L4–S3)	Calf muscles, plantar flexors	Loss of plantar flexion; ankle jerk lost
Superior gluteal (L4–S1)	Gluteus medius and minimus	**Trendelenburg sign** (pelvis drops to the opposite side)

Reflex root values. Biceps C5, C6 · Supinator C5, C6 · Triceps C6, C7 · Knee L3, L4 · Ankle S1, S2. **Hook: "1–2 buckle my shoe (ankle S1–S2), 3–4 kick the door (knee L3–L4), 5–6 pick up sticks (biceps C5–C6), 7–8 lay them straight (triceps C7, with C6)."**

- **Cranial nerves to remember.** Longest: vagus (X).
- Thickest: trigeminal (V).
- Longest intracranial course and the only one leaving the back of the brainstem: trochlear (IV).
- A facial-nerve (VII) lesion at or below its nucleus weakens the whole half-face (Bell's palsy); a lesion above it spares the forehead.
- Hook for the eye muscles: **LR6 SO4** (lateral rectus by VI, superior oblique by IV, the rest by III).

5.2a Cardiovascular physiology

Three things carry most marks: the order of events in the cardiac cycle, the normal ECG intervals, and the baroreceptor and RAAS control of blood pressure.

Cardiac cycle (heart rate 75/min, cycle 0.8 s)

Systole lasts 0.27–0.3 s and diastole about 0.5 s, so diastole shortens most when the heart rate rises.

Phase	Valves	What happens	Duration
1. Atrial systole	AV open, semilunar closed	Atria top up the ventricles (about 20–30% of filling); a-wave in JVP; S4 if the ventricle is stiff	0.1 s
2. Isovolumetric contraction	All four closed	Pressure rises, volume constant; begins with closure of AV valves (**S1**); c-wave in JVP	0.05 s
3. Rapid then reduced ejection	Semilunar open	Blood leaves; aortic pressure peaks at about 120 mmHg	0.22 s
4. Isovolumetric relaxation	All four closed	Begins with closure of aortic and pulmonary valves (**S2**); v-wave builds in the atria	0.08 s

Phase	Valves	What happens	Duration
5. Rapid then slow filling (diastasis)	AV open	Passive filling, about 70% of filling; **S3** in rapid filling (normal in children and athletes)	about 0.4 s

Key numbers. End-diastolic volume 120–130 mL; end-systolic volume 50–60 mL; stroke volume about 70 mL; ejection fraction (SV/EDV) 55–70%; cardiac output = SV × HR = about 5 L/min; cardiac index about 3.2 L/min/m².

- **Heart sounds.** S1 "lub" = closure of mitral and tricuspid valves, start of systole.
- S2 "dub" = closure of aortic and pulmonary valves, end of systole; it splits on inspiration (physiological splitting).
- S3 = rapid ventricular filling.
- S4 = atrial contraction into a stiff ventricle (always abnormal).

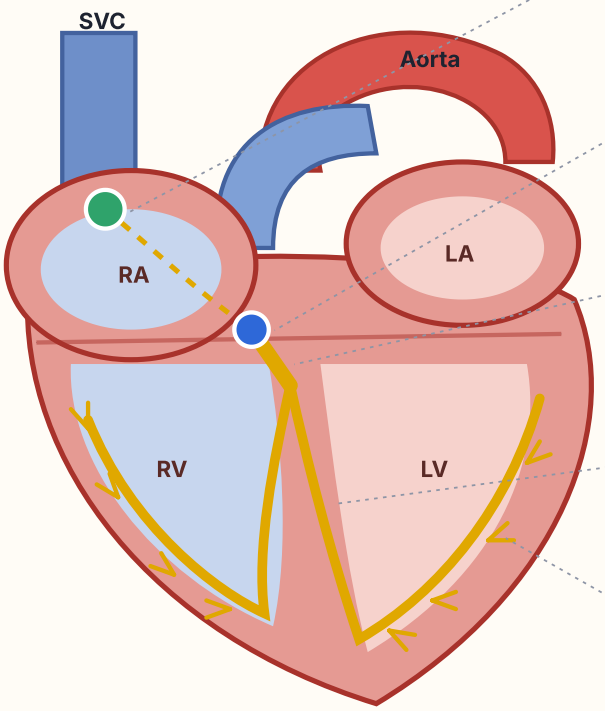
Laws. Frank–Starling: the greater the stretch (EDV), the stronger the contraction. Laplace: wall tension = pressure × radius / (2 × wall thickness), so a dilated heart works harder.

Conducting system and ECG

SA node (pacemaker, 70–80/min) → atria → AV node (delay of about 0.1 s, so atria empty first; intrinsic rate 40–60/min) → bundle of His → right and left bundle branches → Purkinje fibres (fastest conduction, 4 m/s; intrinsic rate 20–40/min).

How the heartbeat spreads

The conducting system, from pacemaker to ventricular wall



1 **SA node**
The pacemaker: 70–80/min

2 **AV node**
Delays about 0.1 s so the atria empty first; own rate 40–60/min

3 **Bundle of His**
Through the septum

4 **Right and left bundle branches**
Down each side of the septum

5 **Purkinje fibres**
Fastest conduction, 4 m/s; own rate 20–40/min

Front view, chambers opened. The impulse runs top to bottom: SA → AV → His → branches → Purkinje.

Conducting system · SA node → AV node → His → bundle branches → Purkinje

ECG item	Represents	Normal
P wave	Atrial depolarisation	Up to 0.1 s (2.5 small squares)
PR interval	Atrial depolarisation + AV-node delay	0.12–0.20 s (3–5 small squares)
QRS complex	Ventricular depolarisation (atrial repolarisation is hidden in it)	0.08–0.10 s, under 0.12 s
ST segment	Ventricles fully depolarised (plateau)	Isoelectric
T wave	Ventricular repolarisation	Upright in most leads
QT interval	Whole of ventricular depolarisation and repolarisation	About 0.40 s; corrected QTc under 0.44 s
U wave	Repolarisation of papillary muscles or Purkinje fibres	Prominent in hypokalaemia

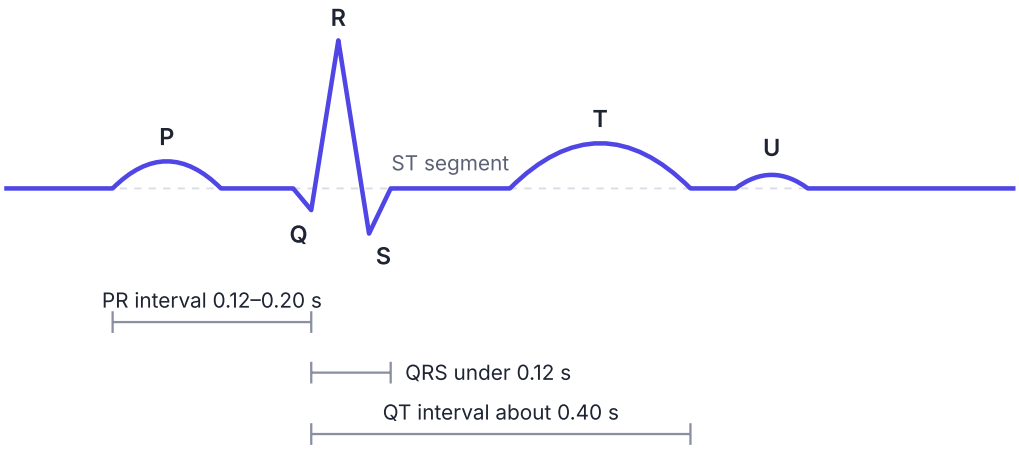
Paper. At 25 mm/s one small square = 0.04 s and one large square = 0.2 s; 1 mV = 10 mm. Heart rate = 300 ÷ number of large squares between two R waves (or 1500 ÷ small squares). Standard 12 leads: 3 bipolar limb leads (I, II, III; Einthoven's triangle), 3 augmented unipolar limb leads (aVR, aVL, aVF), 6 chest leads (V1–V6).

Change	Seen in
Tall peaked T, wide QRS	Hyperkalaemia
Flat T, prominent U wave, ST depression	Hypokalaemia
ST elevation in the leads facing the infarct	Acute myocardial infarction
Prolonged PR interval	First-degree heart block
Short QT / long QT	Hypercalcaemia / hypocalcaemia

Memory hook: "P for Pump-up of atria, QRS for the big ventricles, T for Tidy-up (repolarisation)."

One heartbeat on the ECG: P, QRS and T in order

Lead II, normal sinus beat (schematic) with the normal intervals



P wave = atria depolarise (contract)
QRS complex = ventricles depolarise; atrial repolarisation is hidden in it
T wave = ventricles repolarise · U wave = prominent in low potassium

At 25 mm/s: 1 small square = 0.04 s, 1 large square = 0.2 s; heart rate = 300 ÷ large squares between two R waves

ECG · one normal beat with its waves and intervals

The PR interval includes the AV-node delay; the QT interval covers the whole of ventricular activity, depolarisation plus repolarisation.

Regulation of blood pressure

Mean arterial pressure = diastolic pressure + one third of pulse pressure (about 93 mmHg at 120/80). Blood pressure = cardiac output × total peripheral resistance; resistance is set mainly by the arterioles (Poiseuille: flow varies with the fourth power of the radius).

Control	Speed	How it works
Baroreceptor reflex	Seconds	Stretch receptors in the carotid sinus (via glossopharyngeal, IX) and aortic arch (via vagus, X) → vasomotor centre in the medulla. Rise in BP → more vagal tone, less sympathetic tone → slower heart, vasodilatation. Works best between 60 and 180 mmHg.
Chemoreceptor reflex	Seconds	Carotid and aortic bodies respond to low O_2, high CO_2, low pH; raise BP and breathing when BP falls below about 80 mmHg
CNS ischaemic response	Seconds	Powerful sympathetic discharge when the brain is ischaemic (Cushing reflex: high BP, slow pulse with raised intracranial pressure)
Renin–angiotensin–aldosterone system	Minutes to hours	Low renal perfusion → renin from juxtaglomerular cells → angiotensinogen to angiotensin I → ACE (mostly in lungs) → angiotensin II: vasoconstriction, aldosterone (salt and water retention), ADH, thirst
ADH (vasopressin)	Minutes	From the posterior pituitary on high osmolality or low volume; water retention and vasoconstriction
Atrial natriuretic peptide	Hours	Released when the atria stretch; sodium and water loss, vasodilatation
Renal–body fluid mechanism (pressure natriuresis)	Days (long-term)	The kidneys excrete salt and water until BP returns to its set point; the dominant long-term control

Memory hook: "Baroreceptors are the Buffer, the Kidneys are the Keeper." Baroreceptors smooth out moment-to-moment swings; long-term BP is set by the kidneys.

5.2b Respiratory physiology

The exam asks the muscles of breathing, the lung volumes, surfactant and compliance, and how O_2 and CO_2 are carried.

Mechanics of breathing

- **Inspiration is active.** The diaphragm (phrenic nerve, C3–C5: "C3, 4, 5 keeps the diaphragm alive") does about 75% of quiet breathing; external intercostals lift the ribs. Forced inspiration adds sternocleidomastoid, scalenes and pectorals.
- **Quiet expiration is passive**, from elastic recoil of the lungs and chest wall. Forced expiration uses the internal intercostals and abdominal muscles.
- **Pressures.** Intrapleural pressure is about −2.5 to −3 mmHg at the end of expiration and about −6 mmHg at the end of inspiration; it keeps the lungs expanded. Air enters because alveolar pressure falls below atmospheric (about −1 mmHg).
- **Compliance** = change in volume ÷ change in pressure (about 200 mL/cmH_2O for lungs alone). Falls in fibrosis and pulmonary oedema; rises in emphysema.
- **Surfactant** (dipalmitoyl phosphatidylcholine), secreted by **type II alveolar cells** from about 24 weeks of gestation, lowers surface tension, keeps small alveoli open (Laplace: P = 2T/r) and prevents oedema. Lack of it in preterm babies = respiratory distress syndrome (hyaline membrane disease).
- **Airway resistance** is highest in the medium-sized bronchi. Sympathetic (β_2) stimulation dilates bronchi; parasympathetic (vagus) constricts them.

Lung volumes and capacities (adult male)

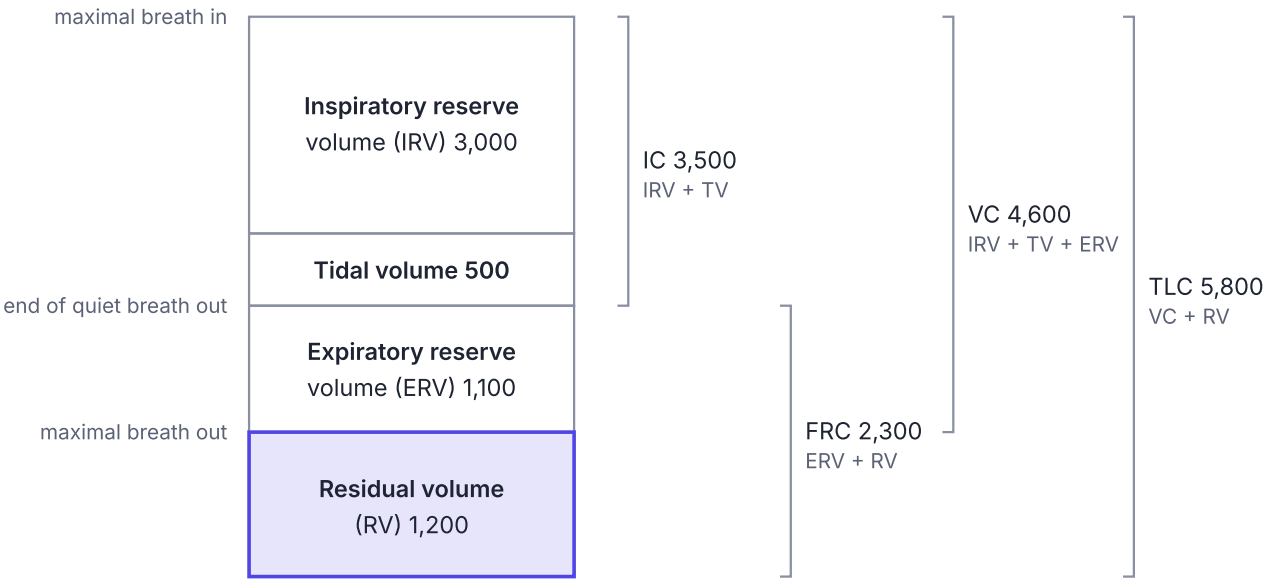
Volume or capacity	Definition	Value
Tidal volume (TV)	Air in or out per quiet breath	500 mL
Inspiratory reserve volume (IRV)	Extra air inhaled after a normal inspiration	3,000 mL
Expiratory reserve volume (ERV)	Extra air exhaled after a normal expiration	1,100 mL
Residual volume (RV)	Air left after maximal expiration; **cannot be measured by spirometry**	1,200 mL
Inspiratory capacity	TV + IRV	3,500 mL
Functional residual capacity (FRC)	ERV + RV; volume at rest	2,300 mL
Vital capacity (VC)	IRV + TV + ERV	4,600 mL

Volume or capacity	Definition	Value
Total lung capacity (TLC)	VC + RV	5,800 mL

Hook: a "capacity" is always two or more volumes added together. RV, FRC and TLC need helium dilution, nitrogen washout or body plethysmography. FEV_1/FVC is normally above 0.75–0.80; it falls in obstructive disease (asthma, COPD) and stays normal or high in restrictive disease (fibrosis).

Capacities are sums of volumes; residual volume always stays in the lungs

Adult male, values in mL (not to scale)



A capacity is two or more volumes added together. Residual volume (highlighted), and so FRC and TLC, cannot be measured with a spirometer: they need helium dilution, nitrogen washout or body plethysmography.

Lung volumes and capacities · 4 volumes, 4 capacities

The four volumes stack from top to bottom; each bracket on the right is a capacity, the sum of the volumes it spans.

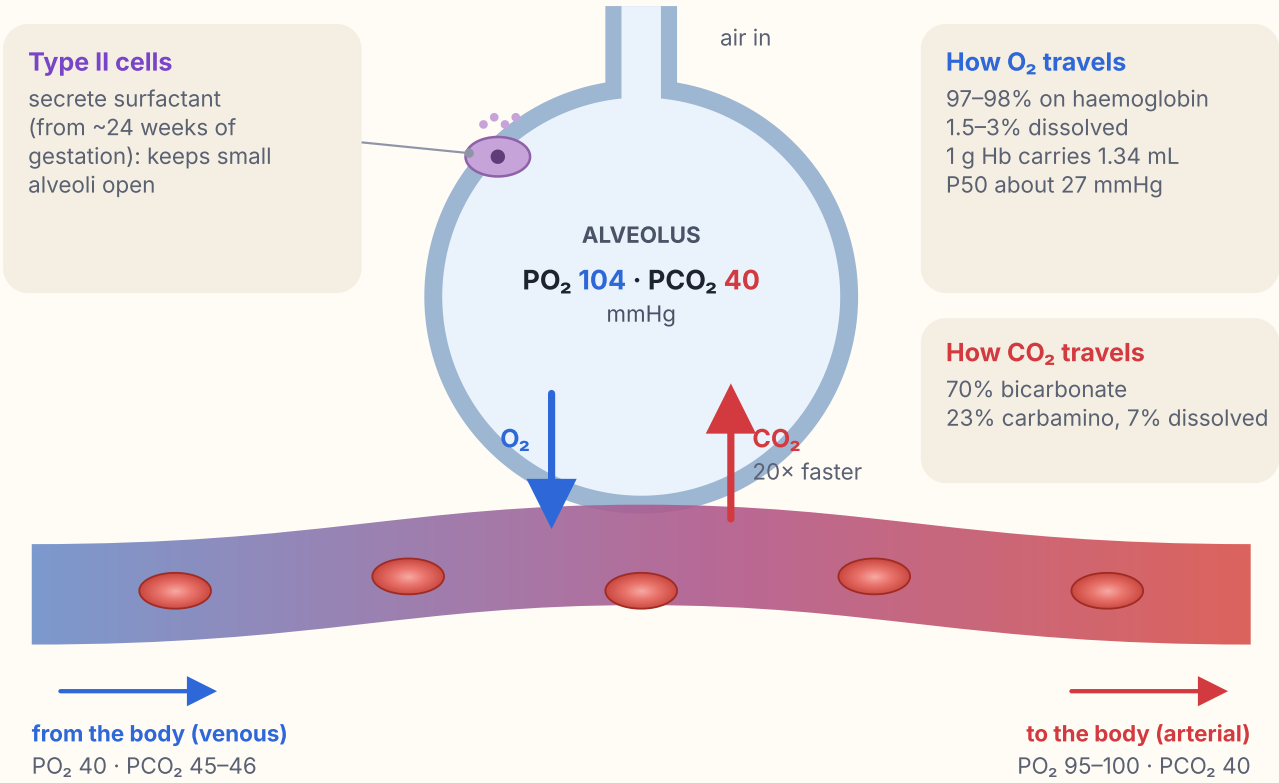
Dead space. Anatomical dead space about 150 mL (about 1 mL per pound of body weight). Alveolar ventilation = (TV − dead space) × rate = (500 − 150) × 12 ≈ 4.2 L/min.

Gas exchange and transport

Item	Value or fact
Alveolar PO_2 / PCO_2	104 / 40 mmHg
Arterial PO_2 / PCO_2	95–100 / 40 mmHg
Mixed venous PO_2 / PCO_2	40 / 45–46 mmHg
O_2 carried	97–98% bound to haemoglobin, 1.5–3% dissolved; 1 g Hb carries 1.34 mL O_2
CO_2 carried	About 70% as bicarbonate (carbonic anhydrase in red cells, chloride shift), 23% as carbamino compounds, 7% dissolved
P50	PO_2 at which Hb is 50% saturated, about 27 mmHg
Diffusion	CO_2 diffuses about 20 times faster than O_2

Gas exchange in the lung

Partial pressures across one alveolus and its capillary (mmHg)



The thin alveolar wall and capillary wall together form the respiratory membrane.

Gas exchange · partial pressures across one alveolus and its capillary

- **Oxygen–haemoglobin dissociation curve** (sigmoid).
- **Shift to the right** (Hb gives up O_2 more easily): raised CO_2, H^+ (low pH: the **Bohr effect**), temperature and 2,3-BPG, as in exercising muscle.

- **Shift to the left**: the opposite, plus fetal haemoglobin and carbon monoxide.
- **Hook: "CADET, face Right"** — CO_2, Acid, 2,3-DPG, Exercise, Temperature shift the curve right.
- The **Haldane effect**: deoxygenated blood carries more CO_2.

Control of breathing

- The dorsal respiratory group in the medulla sets the inspiratory rhythm; the pneumotaxic centre in the pons limits inspiration.
- **Central chemoreceptors** (medulla) respond to H^+ in the CSF, that is, to arterial CO_2: the main moment-to-moment drive.
- **Peripheral chemoreceptors** (carotid bodies via IX, aortic bodies via X) respond mainly to low arterial PO_2 (below about 60 mmHg).
- The **Hering–Breuer reflex** (stretch receptors, vagus) stops over-inflation.

5.2c Renal physiology

Most questions ask the GFR, which substance measures what, and which part of the nephron does what.

Basic numbers

Item	Value
Nephrons per kidney	About 1 million (cortical 85%, juxtamedullary 15%)
Renal blood flow	About 1,200–1,250 mL/min, about 20–25% of cardiac output
Renal plasma flow	About 625–650 mL/min
Glomerular filtration rate (GFR)	About 125 mL/min, or 180 L/day
Filtration fraction (GFR ÷ RPF)	About 0.2 (20%)
Urine output	1–1.5 L/day; 99% of filtrate is reabsorbed
Net filtration pressure	Glomerular capillary pressure 60 − Bowman's capsule pressure 18 − plasma oncotic pressure 32 = about 10 mmHg

Renal clearance

Clearance is the volume of plasma completely cleared of a substance per minute.

$$C = (U \times V) \div P$$

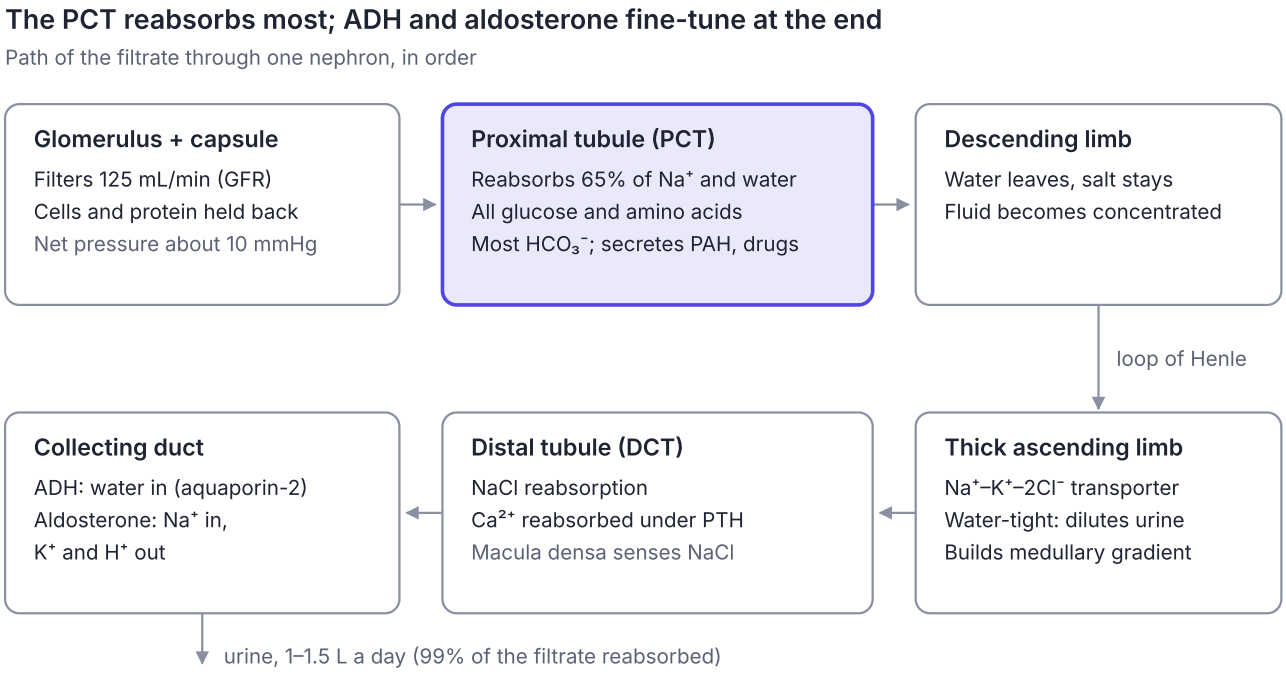
U = urine concentration, V = urine flow (mL/min), P = plasma concentration.

Substance	Handling	Clearance tells you
Inulin	Filtered only; not reabsorbed or secreted	**GFR** (gold standard, 125 mL/min)
Creatinine	Filtered + slightly secreted	GFR in clinical practice (slightly over-estimates)
Para-aminohippuric acid (PAH)	Filtered + almost fully secreted	**Effective renal plasma flow** (about 625 mL/min)
Glucose	Filtered + fully reabsorbed (up to the transport maximum)	Clearance 0; appears in urine above the renal threshold of about 180 mg/dL
Urea	Filtered + partly reabsorbed	About 75 mL/min (less than GFR)

Hook: clearance less than inulin = reabsorbed; more than inulin = secreted.

Nephron, segment by segment

Segment	Main job
Glomerulus and Bowman's capsule	Filtration; proteins and cells held back by the filtration barrier
Proximal convoluted tubule	Reabsorbs about 65% of Na^+ and water, **all glucose and amino acids**, most bicarbonate; secretes drugs and PAH
Descending limb of loop of Henle	Permeable to water, not salt: fluid becomes concentrated
Thick ascending limb	Na^+–K^+–$2Cl^-$ co-transport; impermeable to water ("diluting segment"); builds the medullary gradient (countercurrent multiplier)
Distal convoluted tubule	Na^+–Cl^- reabsorption; calcium reabsorption under PTH
Collecting duct	**Aldosterone**: Na^+ reabsorption, K^+ and H^+ secretion. **ADH**: inserts aquaporin-2, water reabsorption, concentrated urine
Juxtaglomerular apparatus	JG cells (afferent arteriole) secrete renin; the macula densa senses NaCl in the distal tubule



Nephron · what each segment reabsorbs or secretes

Follow the arrows: filtration at the glomerulus, bulk reabsorption in the PCT (highlighted), concentration and dilution in the loop, and hormone-controlled fine-tuning in the distal tubule and collecting duct.

Autoregulation. GFR and renal blood flow stay nearly constant between mean arterial pressures of about 80 and 180 mmHg, through the myogenic response and tubuloglomerular feedback.

Hormones the kidney makes. Renin, erythropoietin (from peritubular interstitial cells; falls in chronic kidney disease, causing anaemia), and calcitriol (1,25-dihydroxy vitamin D, made by 1α-hydroxylase in the proximal tubule).

Micturition. A spinal reflex at S2–S4, controlled by the pons. Parasympathetic (pelvic splanchnic nerves) contracts the detrusor; sympathetic relaxes it; the external sphincter is voluntary (pudendal nerve). The urge is first felt at about 150 mL and becomes marked at 400 mL.

5.3a Gastrointestinal physiology and hepatic functions

Learn one table for digestion (enzyme, source, substrate), one for the GI hormones, the sites of absorption, and the liver's jobs with bilirubin metabolism.

Digestion

Juice (daily volume, pH)	Enzyme	Acts on	Product
Saliva (1–1.5 L, pH 6.8–7)	Salivary amylase (ptyalin)	Starch	Maltose, dextrins
Saliva	Lingual lipase	Fats	Fatty acids (important in infants)
Gastric juice (2–3 L, pH 1–2)	Pepsin (from pepsinogen, activated by HCl)	Proteins	Peptides
Gastric juice	Gastric lipase; rennin (infants)	Fats; milk casein	—
Pancreatic juice (1.5 L, pH 7.8–8.3)	Trypsin, chymotrypsin (trypsinogen activated by enterokinase)	Proteins	Peptides
Pancreatic juice	Pancreatic amylase	Starch	Maltose
Pancreatic juice	Pancreatic lipase + colipase	Triglycerides	Monoglycerides, fatty acids
Bile (0.5–1 L, pH 7.8)	No enzyme; bile salts	Fat	Emulsification and micelles
Intestinal brush border	Maltase, sucrase, lactase, peptidases	Disaccharides, peptides	Monosaccharides, amino acids

Gastric cells. Parietal (oxyntic) cells: HCl and intrinsic factor (needed for vitamin B12; lack = pernicious anaemia). Chief (peptic) cells: pepsinogen. G cells (antrum): gastrin. Mucous neck cells: mucus.

Phases of gastric secretion. Cephalic (sight, smell; vagus; about 30%), gastric (food in the stomach; gastrin; about 60%), intestinal (about 10%).

Gastrointestinal hormones

Hormone	Made by	Stimulus	Main actions
Gastrin	G cells, antrum	Protein, stomach distension, vagus	Acid secretion, gastric motility, mucosal growth
Secretin (first hormone discovered, 1902)	S cells, duodenum	Acid in the duodenum	**Bicarbonate-rich** pancreatic juice; inhibits gastric acid
Cholecystokinin (CCK)	I cells, duodenum and jejunum	Fat and protein	Gall-bladder contraction, **enzyme-rich** pancreatic juice; slows gastric emptying; satiety

Hormone	Made by	Stimulus	Main actions
GIP (glucose-dependent insulinotropic peptide)	K cells	Glucose, fat	Insulin release; inhibits gastric acid
Motilin	M cells	Fasting	Migrating motor complex
Somatostatin	D cells	Acid	Inhibits most GI hormones
Ghrelin	Stomach	Fasting	Hunger

Hook: "Secretin = Soda (bicarbonate); CCK = Contract the gall bladder + Cook (enzymes)."

Absorption

Site	What is absorbed
Stomach	Alcohol, aspirin, little water
Duodenum	**Iron** and **calcium** (active), folate (upper jejunum)
Jejunum	Most carbohydrates, proteins, fats, water-soluble vitamins
Terminal ileum	**Vitamin B12** (with intrinsic factor) and **bile salts** (enterohepatic circulation)
Colon	Water, sodium; vitamin K and biotin made by gut bacteria

Glucose and galactose enter by **SGLT-1** (sodium co-transport, the basis of oral rehydration solution); fructose by facilitated diffusion (GLUT-5). Fats enter as micelles, are rebuilt into chylomicrons and enter the lacteals (lymph), not blood.

Motility. Swallowing: oral (voluntary), pharyngeal and oesophageal (reflex) phases. Peristalsis is run by the myenteric (Auerbach's) plexus; secretion by the submucosal (Meissner's) plexus. Gastrocolic reflex: food in the stomach triggers mass movement in the colon. Defecation reflex: sacral S2–S4.

Hepatic functions

Function	Details
Carbohydrate	Glycogenesis, glycogenolysis, gluconeogenesis: keeps fasting blood glucose steady
Protein	Makes albumin (keeps oncotic pressure), clotting factors (I, II, V, VII, IX, X), urea from ammonia (urea cycle)

Function	Details
Fat	Makes cholesterol, lipoproteins (VLDL), bile salts; beta-oxidation; ketone bodies
Storage	Glycogen, vitamins A, D, B12, iron (ferritin), copper
Detoxification	Drugs, alcohol, ammonia, hormones; conjugation of bilirubin
Bile	Bile salts, bilirubin, cholesterol; about 600 mL/day
Immune	Kupffer cells (macrophages) remove bacteria from portal blood
Fetal blood	The liver makes red cells in fetal life

Vitamin K-dependent factors: "1972" = II, VII, IX, X (plus proteins C and S). A raised prothrombin time is an early sign of poor liver synthesis; low albumin a late one (half-life about 20 days).

Bilirubin metabolism. Old red cells → haem → biliverdin → **unconjugated bilirubin** (fat-soluble, carried on albumin, not in urine) → liver conjugates it with glucuronic acid (UDP-glucuronyl transferase) → **conjugated bilirubin** (water-soluble) → bile → gut bacteria make **urobilinogen** → stercobilin (colours stool brown) and some reabsorbed and excreted in urine as urobilin. Normal serum bilirubin 0.3–1.2 mg/dL; jaundice shows above about 2–2.5 mg/dL.

Jaundice	Raised bilirubin	Urine	Stool
Pre-hepatic (haemolytic)	Unconjugated	No bilirubin; urobilinogen raised	Dark
Hepatic (hepatitis)	Both	Bilirubin present	Normal or pale
Post-hepatic (obstructive)	Conjugated	Bilirubin present (dark); urobilinogen absent	**Clay-coloured**, with itching

5.3b Endocrine: hypothalamic-pituitary axis and feedback loops

The hypothalamus sends releasing hormones down the portal veins to the anterior pituitary, the anterior pituitary drives the target glands, and the target-gland hormone switches both off. That loop, negative feedback, answers most questions.

Hypothalamus → anterior pituitary → target

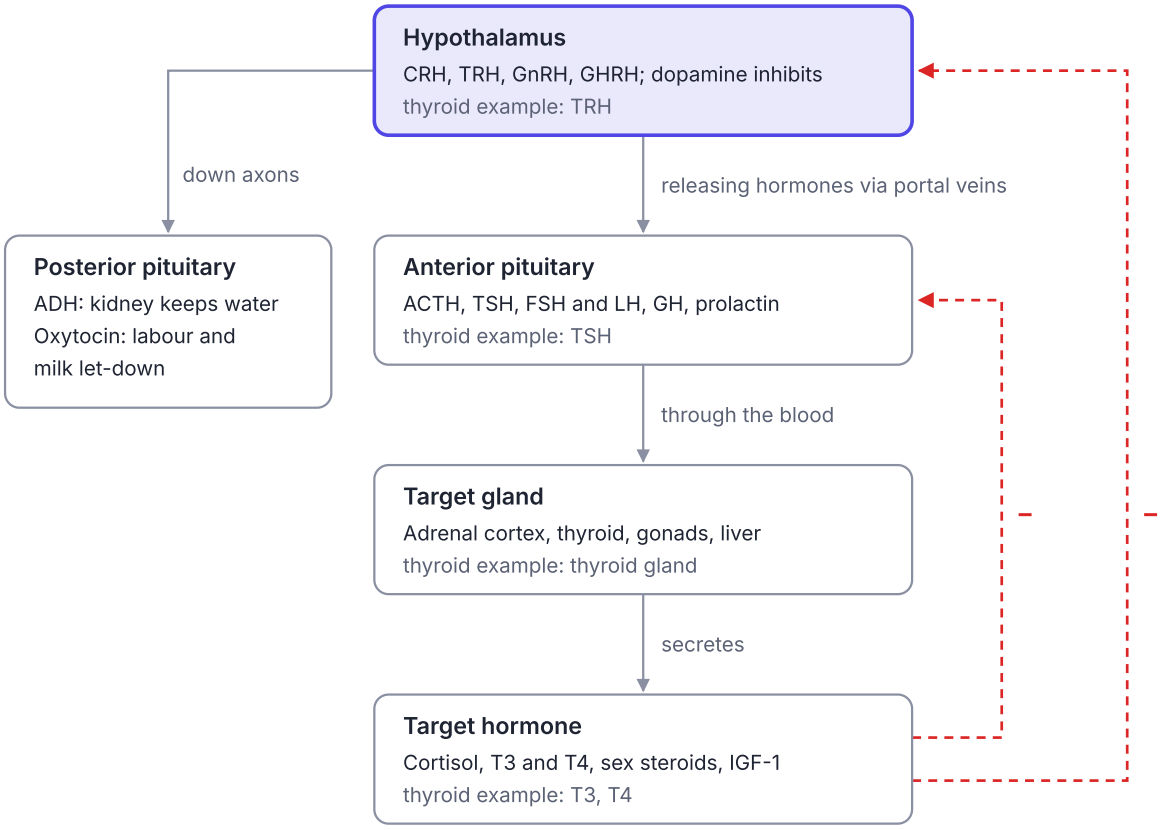
Hypothalamic hormone	Anterior pituitary hormone	Target and effect	Feedback
TRH	TSH	Thyroid: T3, T4 (metabolic rate)	T3/T4 inhibit TRH and TSH
CRH	ACTH	Adrenal cortex: cortisol	Cortisol inhibits CRH and ACTH
GnRH (pulsatile)	FSH, LH	Gonads: oestrogen, progesterone, testosterone; follicles, sperm	Sex steroids, inhibin
GHRH (stimulates); somatostatin (inhibits)	Growth hormone	Liver: IGF-1; growth, protein synthesis, raises glucose	IGF-1, GH
Dopamine (inhibits); TRH (stimulates)	Prolactin	Breast: milk production	**Prolactin is the only pituitary hormone mainly under inhibitory control**

- **Posterior pituitary (neurohypophysis).** Stores hormones made in the hypothalamus (supraoptic and paraventricular nuclei) and sent down axons.
- **ADH (vasopressin)**: water reabsorption in the collecting duct; lack = diabetes insipidus (large volumes of dilute urine).
- **Oxytocin**: milk ejection (let-down) and uterine contraction in labour.

Feedback types. Negative feedback (most hormones): the product switches off its own drive. **Positive feedback** (rare): the LH surge before ovulation (rising oestrogen), oxytocin in labour (Ferguson reflex), clotting cascade.

The target hormone switches off its own drive (negative feedback)

Hypothalamic-pituitary axis, with the thyroid as the worked example



Red dashed arrows: the target hormone inhibits both the pituitary and the hypothalamus (negative feedback). Exceptions with positive feedback: the LH surge before ovulation and oxytocin in labour.

Hypothalamic-pituitary axis · 4 levels and 2 feedback loops

In hypothyroidism from a failing thyroid, T4 falls, feedback is lost, and TSH rises; that is why a high TSH is the first sign of primary hypothyroidism.

Other glands at a glance

Gland	Hormone	Key action	Too little / too much
Thyroid follicular cells	T4 (mostly), T3 (more active)	Basal metabolic rate, growth, brain development	Cretinism, myxoedema / Graves' disease
Thyroid parafollicular (C) cells	Calcitonin	Lowers blood calcium	—
Parathyroid (chief cells)	Parathormone (PTH)	Raises blood calcium (bone resorption, kidney reabsorption, activates vitamin D)	Tetany (after thyroid surgery) / bone loss, kidney stones

Gland	Hormone	Key action	Too little / too much
Adrenal cortex: zona glomerulosa	Aldosterone (mineralocorticoid)	Na^+ retention, K^+ loss	Addison's disease / Conn's syndrome
Zona fasciculata	Cortisol (glucocorticoid)	Raises glucose, anti-inflammatory, stress	Addison's / Cushing's syndrome
Zona reticularis	Androgens (DHEA)	Pubic and axillary hair	—
Adrenal medulla	Adrenaline (80%), noradrenaline	Fight or flight	— / phaeochromocytoma
Pancreas: β cells (about 70%)	Insulin	Lowers glucose	Diabetes mellitus / hypoglycaemia
Pancreas: α cells	Glucagon	Raises glucose	—
Pancreas: δ cells	Somatostatin	Inhibits insulin and glucagon	—
Pineal	Melatonin	Sleep–wake rhythm	—

Adrenal cortex hook: "Salt, Sugar, Sex — the deeper you go, the sweeter it gets" (glomerulosa, fasciculata, reticularis: "GFR"). The cortex comes from mesoderm, the medulla from neural crest.

Thyroid facts. Iodide is trapped by the sodium–iodide symporter, oxidised by thyroid peroxidase and attached to thyroglobulin. T4 is the main product; T3 is 3–5 times more potent and is mostly made from T4 in the tissues. Normal TSH is about 0.4–4.5 mIU/L.

Calcium. Normal serum calcium 9–11 mg/dL (about half ionised). PTH and vitamin D raise it; calcitonin lowers it.

Memory hook: the adrenal cortex from outside in is **salt, sugar, sex**: zona **G**lomerulosa makes aldosterone (salt), **F**asciculata cortisol (sugar), **R**eticularis androgens (sex), "**GFR**". The deeper you go, the sweeter it gets.

5.4a Body fluids, electrolytes and acid-base balance

The numbers to learn are the 60–40–20 rule, the main ion inside and outside the cell, and normal arterial blood gas values; acid-base questions are solved by the four-step flowchart below.

Fluid compartments (70 kg adult male)

Compartment	Share of body weight	Volume	Measured with
Total body water	60%	42 L	Heavy water (D_2O), antipyrine
Intracellular fluid (ICF)	40% (two-thirds of TBW)	28 L	TBW − ECF
Extracellular fluid (ECF)	20% (one-third of TBW)	14 L	Inulin, mannitol, sucrose
— Interstitial fluid	15%	about 10.5 L	ECF − plasma
— Plasma	5%	about 3.5 L	Evans blue (T-1824), radio-iodinated albumin
Blood volume	about 7–8%	5 L	Plasma volume ÷ (1 − haematocrit)

Hook: "60–40–20" (total, inside, outside). Women and the elderly have less water (more fat); infants more (about 75%).

Ion	ECF (mEq/L)	ICF (mEq/L)
Na^+	142 (main ECF cation)	10–14
K^+	4	140 (main ICF cation)
Cl^-	103 (main ECF anion)	4
HCO_3^-	24–28	10
Phosphate, proteins	low	high (main ICF anions)

The Na^+–K^+ ATPase (3 Na^+ out, 2 K^+ in per ATP) keeps this difference. Normal plasma osmolality is 285–295 mOsm/kg; estimated as 2 × Na^+ + glucose/18 + urea/2.8 (mg/dL). 0.9% NaCl is isotonic with plasma.

Starling forces decide movement between plasma and interstitium: capillary hydrostatic pressure pushes fluid out; plasma oncotic pressure (albumin, about 25–28 mmHg) pulls it in. Oedema follows high venous pressure (heart failure), low albumin (nephrotic syndrome, liver disease), leaky capillaries (inflammation) or blocked lymphatics (filariasis).

Water balance. Daily intake about 2.5 L (drink 1.5 L, food 0.8 L, metabolic water 0.3 L) equals output (urine 1.5 L, skin and lungs about 0.9 L as insensible loss, faeces 0.1 L). Thirst and ADH respond to a 1–2% rise in osmolality.

Acid-base balance

Arterial blood	Normal
pH	7.35–7.45 (7.40)
$PaCO_2$	35–45 mmHg (40)
HCO_3^-	22–26 mEq/L (24)
Anion gap = $Na^+ - (Cl^- + HCO_3^-)$	8–16 mEq/L (about 12)

Henderson–Hasselbalch.

$$pH = 6.1 + \log\left([HCO_3^-] \div (0.03 \times PaCO_2) \right)$$

The ratio of bicarbonate to dissolved CO_2 is 20 : 1 at pH 7.4.

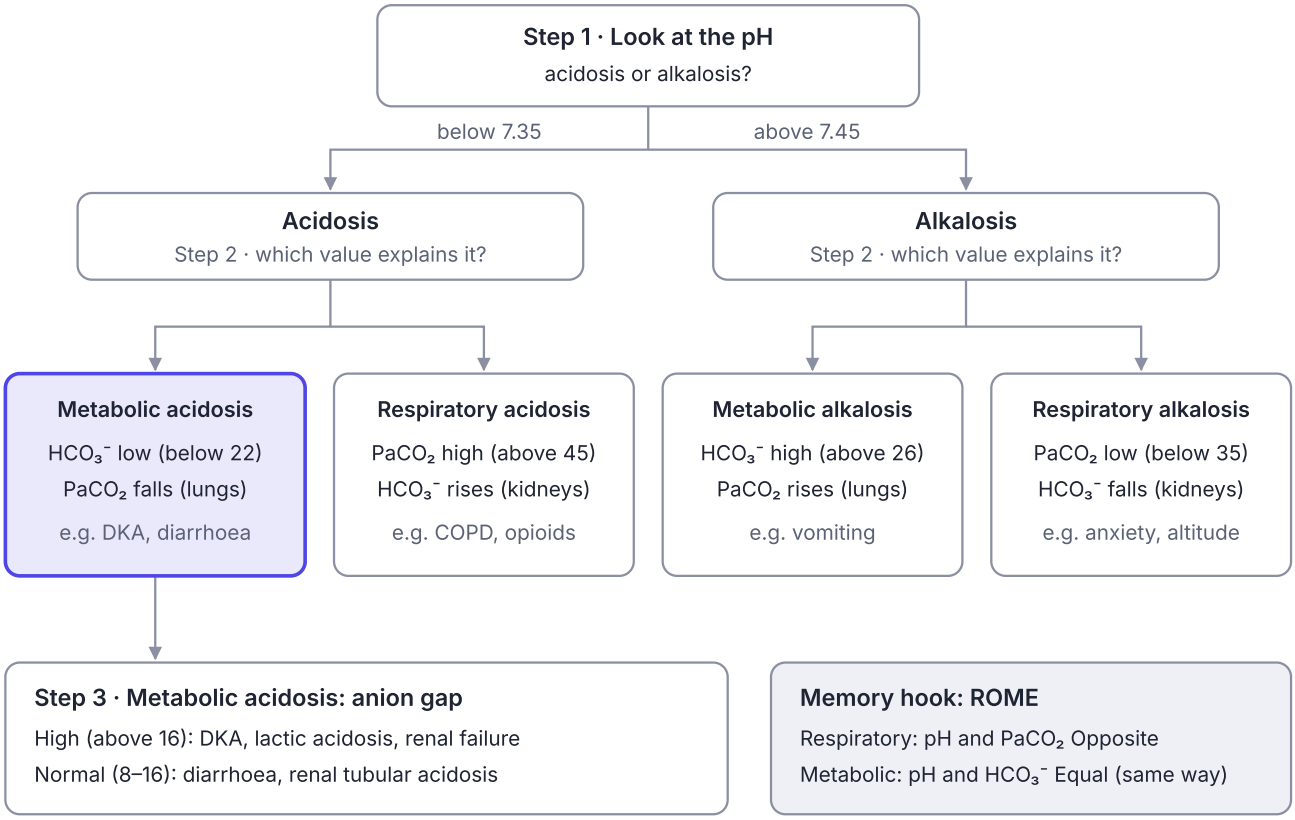
Buffers. Bicarbonate is the main buffer of the ECF; haemoglobin of the blood (red cells); phosphate and proteins inside cells, phosphate also in urine. Lines of defence: buffers (seconds), lungs (minutes, by changing CO_2), kidneys (days, by excreting H^+ as ammonium and titratable acid and making new HCO_3^-).

Disorder	Primary change	Compensation	Common causes
Metabolic acidosis	HCO_3^- falls	Hyperventilation, $PaCO_2$ falls (Kussmaul breathing)	Diabetic ketoacidosis, lactic acidosis, renal failure (high anion gap); diarrhoea (normal anion gap)
Metabolic alkalosis	HCO_3^- rises	Hypoventilation, $PaCO_2$ rises	Vomiting, nasogastric suction, diuretics, hypokalaemia
Respiratory acidosis	$PaCO_2$ rises	Kidneys keep HCO_3^-	COPD, opioid overdose, chest-wall weakness
Respiratory alkalosis	$PaCO_2$ falls	Kidneys excrete HCO_3^-	Anxiety hyperventilation, high altitude, early salicylate poisoning

Hook: "ROME" — Respiratory Opposite, Metabolic Equal. In respiratory disorders pH and $PaCO_2$ move in opposite directions; in metabolic disorders pH and HCO_3^- move in the same direction. High-anion-gap metabolic acidosis causes: **"MUDPILES"** (methanol, uraemia, diabetic ketoacidosis, paraldehyde, iron or isoniazid, lactic acidosis, ethylene glycol, salicylates).

Three steps to any blood gas: pH, cause, anion gap

Normal: pH 7.35–7.45 · $PaCO_2$ 35–45 mmHg · HCO_3^- 22–26 mEq/L



Acid-base flowchart · 3 steps to the primary disorder

Worked example: pH 7.25, $PaCO_2$ 30, HCO_3^- 14. The pH is low (acidosis); HCO_3^- is low, which explains it (metabolic); $PaCO_2$ is low because the lungs are compensating.

Key electrolytes

Ion	Normal serum	Low	High
Sodium	135–145 mEq/L	Confusion, seizures (SIADH, diuretics)	Thirst, confusion (water loss, diabetes insipidus)
Potassium	3.5–5.0 mEq/L	Muscle weakness, flat T and U waves	Peaked T waves, cardiac arrest

Ion	Normal serum	Low	High
Calcium	9–11 mg/dL	Tetany: Chvostek's and Trousseau's signs	Stones, bones, groans, moans
Chloride	96–106 mEq/L	Metabolic alkalosis (vomiting)	Normal anion gap acidosis
Magnesium	1.5–2.5 mg/dL	Tetany, arrhythmia	Drowsiness, loss of reflexes

5.4b Clinical biochemistry: enzymes, lipid profile, blood glucose

For each test, know the normal range and the organ a raised value points to.

Enzymes

- **Basics.** Enzymes are protein catalysts that lower activation energy; they do not change the equilibrium.
- **Km** (Michaelis constant) = substrate concentration at half the maximum velocity; a low Km means high affinity.
- Competitive inhibitors raise Km, Vmax unchanged; non-competitive inhibitors lower Vmax, Km unchanged.
- Isoenzymes catalyse the same reaction but differ in structure (for example, LDH and CK).

Six classes (IUBMB): "Over The HILL" — Oxidoreductases, Transferases, Hydrolases, Isomerases, Lyases, Ligases.

Enzyme	Normal (approx.)	Raised in
ALT (SGPT)	7–56 U/L	Liver cell injury; **more specific for the liver**
AST (SGOT)	10–40 U/L	Liver, heart, muscle; AST/ALT above 2 suggests alcoholic liver disease
Alkaline phosphatase (ALP)	40–130 U/L	Obstructive jaundice, bone disease (rickets, Paget's), normal growth and pregnancy
Gamma-glutamyl transferase (GGT)	under 50 U/L	Alcohol, cholestasis (confirms a liver source of raised ALP)
CK-MB	under 5% of total CK	Myocardial infarction: rises 4–6 h, peaks 24 h, normal in 2–3 days

Enzyme	Normal (approx.)	Raised in
Cardiac troponin I or T	below assay cut-off	Myocardial infarction: most specific; stays up 7–10 days
LDH (LDH-1 > LDH-2 "flipped")	140–280 U/L	Late marker of MI; haemolysis
Amylase and lipase	Amylase 30–110 U/L	Acute pancreatitis (lipase more specific, stays raised longer)
Acid phosphatase (prostatic)	—	Prostate cancer (now replaced by PSA)

Hook for liver tests: "ALT = Liver (L for Liver); ALP = Pipes (obstruction, bile ducts and bone)."

Lipid profile (fasting 9–12 hours)

Test	Desirable	High
Total cholesterol	under 200 mg/dL	240 mg/dL or more
LDL cholesterol ("bad")	under 100 mg/dL	160 mg/dL or more
HDL cholesterol ("good")	40 mg/dL or more (men), 50 or more (women); 60 or more protective	—
Triglycerides	under 150 mg/dL	200 mg/dL or more
VLDL	about TG ÷ 5	—

Friedewald formula: LDL = total cholesterol − HDL − (triglycerides ÷ 5), in mg/dL, valid when triglycerides are below 400.

Lipoprotein	Carries	From	Note
Chylomicrons	Dietary triglycerides	Intestine	Largest, least dense; make fasting serum milky
VLDL	Liver-made triglycerides	Liver	Becomes IDL, then LDL

Lipoprotein	Carries	From	Note
LDL	Cholesterol to tissues	From VLDL	Atherogenic; taken up by LDL receptors (apo B-100)
HDL	Cholesterol back to the liver (reverse transport)	Liver, intestine	Smallest, densest; protective (apo A-I)

Hook: "HDL = Healthy, LDL = Lethal." HMG-CoA reductase is the rate-limiting enzyme of cholesterol synthesis.

Blood glucose regulation

Test	Normal	Prediabetes	Diabetes (ADA, WHO)
Fasting plasma glucose	70–99 mg/dL	100–125 mg/dL	126 mg/dL or more
2-hour glucose after 75 g oral glucose	under 140 mg/dL	140–199 mg/dL	200 mg/dL or more
HbA1c	under 5.7%	5.7–6.4%	6.5% or more
Random plasma glucose with symptoms	—	—	200 mg/dL or more

Hypoglycaemia: below 70 mg/dL (symptoms usually below about 55). The renal threshold for glucose is about 180 mg/dL. HbA1c reflects the average glucose of the past 2–3 months (red-cell lifespan 120 days).

Hormone	Effect on glucose	How
Insulin (the only hormone that **lowers** glucose)	Lowers	Glucose uptake into muscle and fat (GLUT-4), glycogen synthesis, fat storage; stops gluconeogenesis
Glucagon	Raises	Glycogenolysis and gluconeogenesis in the liver
Adrenaline	Raises	Glycogenolysis in liver and muscle (fast stress response)
Cortisol	Raises	Gluconeogenesis
Growth hormone	Raises	Opposes insulin action

- **Fed and fasting states.** After a meal: insulin high, glycogen stored.
- Overnight fast: liver glycogenolysis keeps glucose steady.

- Longer fasting (about 24 hours and beyond): gluconeogenesis from amino acids, lactate and glycerol; after several days, ketone bodies feed the brain.
- Liver glycogen lasts about 12–18 hours.
- Glucose enters the brain and red cells without insulin (GLUT-1, GLUT-3).

Rapid revision: normal values and one-line facts

Read this sheet every day until the exam; most Topic 5 questions are one of these lines.

System	Value or fact
Blood	Hb 13–17 g/dL (men), 12–15 (women); RBC 4.5–5.5 million/µL; WBC 4,000–11,000/µL; platelets 1.5–4 lakh/µL; RBC lifespan 120 days; platelet 7–10 days
Blood	Universal donor O negative; universal recipient AB positive; ESR (Westergren) men 0–15, women 0–20 mm/h
Heart	HR 60–100/min; BP 120/80 mmHg; cardiac output 5 L/min; SV 70 mL; EF 55–70%
Heart	PR 0.12–0.20 s; QRS under 0.12 s; QT about 0.40 s; SA node = pacemaker; AV node = delay
Lungs	TV 500 mL; VC 4.6 L; TLC 5.8 L; RV 1.2 L (not by spirometry); dead space 150 mL; RR 12–16/min
Lungs	Surfactant from type II pneumocytes; O_2–Hb curve shifts right with CO_2, acid, 2,3-BPG, exercise, heat
Kidney	GFR 125 mL/min (inulin); RPF 625 mL/min (PAH); RBF 1.2 L/min; filtration fraction 20%; renal glucose threshold 180 mg/dL
Kidney	Glucose and amino acids fully reabsorbed in the PCT; ADH acts on the collecting duct; renin from JG cells
GI	Gastric pH 1–2; HCl and intrinsic factor from parietal cells; B12 and bile salts absorbed in the terminal ileum; iron in the duodenum
GI	Secretin = bicarbonate juice; CCK = gall-bladder contraction and enzymes; gastrin = acid
Liver	Bilirubin 0.3–1.2 mg/dL; albumin 3.5–5 g/dL; vitamin K factors II, VII, IX, X; Kupffer cells = macrophages
Endocrine	Prolactin under dopamine inhibition; ADH and oxytocin made in the hypothalamus, stored in the posterior pituitary
Endocrine	Adrenal cortex: glomerulosa aldosterone, fasciculata cortisol, reticularis androgens; insulin from β cells
Fluids	TBW 60% of body weight; ICF 40%; ECF 20%; plasma 5% (Evans blue); main ICF cation K^+, main ECF cation Na^+

System	Value or fact
Acid-base	pH 7.35–7.45; $PaCO_2$ 35–45 mmHg; HCO_3^- 22–26 mEq/L; anion gap 8–16; HCO_3^- : dissolved CO_2 = 20 : 1
Electrolytes	Na^+ 135–145; K^+ 3.5–5.0 mEq/L; Ca^{2+} 9–11 mg/dL; serum osmolality 285–295 mOsm/kg
Biochemistry	Fasting glucose 70–99 mg/dL; diabetes fasting 126 or more, HbA1c 6.5% or more; total cholesterol under 200; LDL under 100; HDL 40 or more; TG under 150
Biochemistry	ALT liver-specific; ALP raised in obstruction and bone disease; troponin most specific for MI
Anatomy	Spinal cord ends at L1 (adult); lumbar puncture at L3–L4; sternal angle = 2nd rib, T4–T5
Anatomy	Wrist drop = radial; claw hand = ulnar; ape thumb = median; foot drop = common peroneal; winged scapula = long thoracic
Anatomy	Broca = motor speech (frontal); Wernicke = understanding (temporal); vision = occipital (area 17); cerebellum = same-side ataxia

One-glance recall card

Topic 5 recall card: the 15 facts asked most

Cover the small lines, say the fact from the big cue, then check

Broca 44, 45
motor speech, frontal;
Wernicke 22, temporal

LR6 SO4
lateral rectus VI,
superior oblique IV

C5 to T1
roots of the
brachial plexus

PR 0.12-0.20 s
QRS under 0.12 s;
1 small square = 0.04 s

CADET, face right
CO2, acid, DPG, exercise,
temperature shift O2 curve

TV 500 mL
tidal volume; VC is
IRV + TV + ERV

GFR 125
mL/min; inulin clearance
measures it

PAH clearance
measures renal
plasma flow

Parietal cells
acid and intrinsic
factor (B12)

Glucose: SGLT-1
sodium-linked uptake
in the small gut

Salt, sugar, sex
adrenal cortex: glomerulosa,
fasciculata, reticularis

60 · 40 · 20
body water %, ICF %, ECF %
of body weight

20 : 1
bicarbonate to CO2
keeps pH at 7.4

MUDPILES
causes of high anion
gap acidosis

LDL = TC-HDL-TG/5
Friedewald formula,
mg/dL

Topic 5 recall card · 15 tiles

For the last ten minutes before the exam: cover the small lines of each tile and say the fact from the big cue.

Practice MCQs

Thirty questions in the CBRT style, grouped by syllabus heading; the answer key with a one-line reason follows. Option E in the real paper is always "None of the above".

Neuro-musculoskeletal anatomy

1. The first carpometacarpal joint of the thumb is a: (a) hinge joint (b) saddle joint (c) pivot joint (d) plane joint
2. Expressive (non-fluent) aphasia follows a lesion of: (a) Wernicke's area (b) Broca's area (c) angular gyrus (d) occipital lobe
3. Wrist drop after a fracture of the shaft of the humerus is due to injury of the: (a) median nerve (b) ulnar nerve (c) radial nerve (d) axillary nerve
4. The pubic symphysis is an example of a: (a) primary cartilaginous joint (b) secondary cartilaginous joint (c) syndesmosis (d) synovial joint
5. Trendelenburg's sign is positive in paralysis of the: (a) gluteus maximus (b) gluteus medius and minimus (c) quadriceps (d) hamstrings
6. The knee jerk tests spinal segments: (a) S1, S2 (b) L3, L4 (c) C5, C6 (d) C7, C8

Cardiovascular, respiratory and renal

7. The first heart sound is caused by closure of the: (a) aortic and pulmonary valves (b) mitral and tricuspid valves (c) mitral and aortic valves (d) pulmonary valve only
8. The PR interval on the ECG mainly represents: (a) ventricular repolarisation (b) atrial depolarisation and the delay in the AV node (c) ventricular depolarisation (d) atrial repolarisation
9. The main long-term regulator of arterial blood pressure is the: (a) baroreceptor reflex (b) chemoreceptor reflex (c) renal–body fluid mechanism (d) CNS ischaemic response
10. The lung volume that cannot be measured with a simple spirometer is the: (a) tidal volume (b) vital capacity (c) residual volume (d) inspiratory reserve volume
11. Pulmonary surfactant is secreted by: (a) type I pneumocytes (b) type II pneumocytes (c) alveolar macrophages (d) Clara cells
12. A shift of the oxygen–haemoglobin dissociation curve to the right is caused by: (a) fall in temperature (b) fetal haemoglobin (c) rise in PCO_2 (d) carbon monoxide

13. The clearance of which substance measures the glomerular filtration rate most accurately? (a) PAH (b) urea (c) inulin (d) glucose
14. Glucose is normally reabsorbed almost completely in the: (a) proximal convoluted tubule (b) loop of Henle (c) distal convoluted tubule (d) collecting duct

Gastrointestinal, hepatic and endocrine

15. Intrinsic factor is secreted by the: (a) chief cells (b) parietal cells (c) G cells (d) mucous neck cells
16. Vitamin B12 is absorbed mainly in the: (a) duodenum (b) jejunum (c) terminal ileum (d) colon
17. Cholecystokinin causes: (a) bicarbonate-rich pancreatic juice (b) contraction of the gall bladder (c) gastric acid secretion (d) relaxation of the gall bladder
18. Clay-coloured stools with dark urine suggest: (a) haemolytic jaundice (b) obstructive jaundice (c) Gilbert syndrome (d) physiological jaundice
19. The pituitary hormone mainly under inhibitory control of the hypothalamus is: (a) TSH (b) ACTH (c) prolactin (d) growth hormone
20. ADH acts mainly on the: (a) proximal tubule (b) loop of Henle (c) collecting duct (d) glomerulus
21. Aldosterone is secreted by the: (a) zona glomerulosa (b) zona fasciculata (c) zona reticularis (d) adrenal medulla
22. An example of positive feedback is: (a) cortisol suppressing ACTH (b) the LH surge before ovulation (c) T4 suppressing TSH (d) insulin lowering glucose

Body fluids, acid-base and biochemistry

23. Intracellular fluid forms about what share of body weight? (a) 20% (b) 40% (c) 60% (d) 5%
24. The main cation of intracellular fluid is: (a) sodium (b) potassium (c) calcium (d) magnesium
25. Plasma volume is measured with: (a) inulin (b) heavy water (c) Evans blue (d) antipyrine
26. pH 7.25, $PaCO_2$ 30 mmHg and HCO_3^- 14 mEq/L indicate: (a) respiratory acidosis (b) metabolic acidosis with respiratory compensation (c) metabolic alkalosis (d) respiratory alkalosis
27. Persistent vomiting typically produces: (a) metabolic acidosis (b) metabolic alkalosis (c) respiratory acidosis (d) respiratory alkalosis
28. The enzyme most specific for liver-cell injury is: (a) AST (b) ALT (c) ALP (d) LDH
29. Diabetes mellitus is diagnosed when HbA1c is: (a) 5.7% or more (b) 6.0% or more (c) 6.5% or more (d) 7.5% or more
30. The lipoprotein that carries cholesterol from the tissues back to the liver is: (a) chylomicron (b) VLDL (c) LDL (d) HDL

Answer key

Q	Answer	Why
1	b	Saddle joint allows opposition of the thumb
2	b	Broca's area (frontal) = motor speech
3	c	Radial nerve runs in the spiral groove
4	b	Symphysis = secondary cartilaginous, midline
5	b	Superior gluteal nerve palsy; pelvis drops to the opposite side
6	b	Knee jerk L3, L4
7	b	S1 = AV-valve closure at the start of systole
8	b	Includes the AV-node delay; 0.12–0.20 s
9	c	The kidneys set long-term pressure (pressure natriuresis)
10	c	RV stays in the lungs; needs helium dilution or plethysmography
11	b	Type II alveolar cells
12	c	CADET: CO_2, acid, 2,3-BPG, exercise, temperature
13	c	Inulin is filtered, not reabsorbed or secreted
14	a	PCT reabsorbs all glucose up to the transport maximum
15	b	Parietal cells make HCl and intrinsic factor
16	c	B12–intrinsic factor complex in the terminal ileum
17	b	CCK contracts the gall bladder; secretin gives bicarbonate
18	b	Bile cannot reach the gut; conjugated bilirubin spills into urine
19	c	Dopamine inhibits prolactin
20	c	Aquaporin-2 in the collecting duct
21	a	Glomerulosa = salt (mineralocorticoid)
22	b	Rising oestrogen triggers more LH

Q	Answer	Why
23	b	60–40–20 rule
24	b	K^+ about 140 mEq/L inside the cell
25	c	Evans blue binds albumin and stays in plasma
26	b	Low pH, low HCO_3^- (same direction = metabolic); low $PaCO_2$ = compensation
27	b	Loss of gastric HCl
28	b	ALT is mainly in the liver
29	c	6.5% or more (ADA/WHO)
30	d	HDL = reverse cholesterol transport